

Duration: 45 mins

Max Marks: 20

Note: Please write the answers in SI units and up to three decimal points with units. Use $g=9.81 \text{ m/s}^2$.

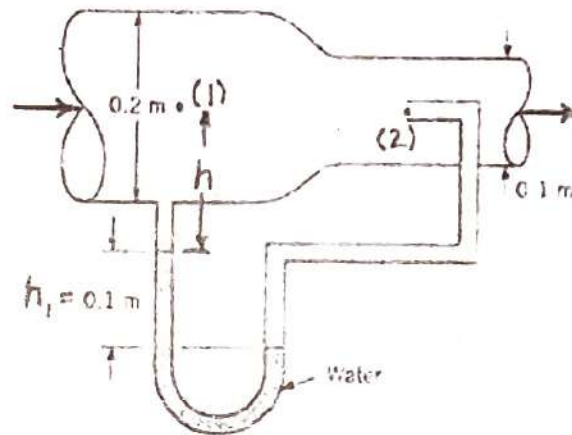
Q1. During a trip to the beach (P_{atm} which is 10.3 m of H_2O), a car runs out of gasoline, and it becomes necessary to siphon gasoline out of the car of a Good Samaritan. Outlet Point 2 is located 0.75 m below point 1 in this case and point 3 is the highest elevation of the siphon which is located 2 m above point 1. The siphon diameter is 5 mm, and frictional losses in the siphon are to be disregarded. Determine (a) the minimum time to withdraw 4 L of gasoline from the tank to the can and (b) the pressure at point 3 (c) Maximum elevation possible for point 3 without causing damage to the tube or choking of flow. The density of gasoline is 750 kg/m^3 .

[6]

Q2. A Pitot-static probe is used to measure the speed of an aircraft flying at 3000 m. If the differential pressure reading is 3 kPa, determine the speed of the aircraft.

[4]

Q3. Air flows steadily through the variable area pipe shown in Fig. Determine the flowrate if viscous and compressibility effects are negligible.



[4]

Q4. Water flows steadily through a variable area vertical pipe downwards. The velocity is given by $\bar{V} = 10(1 + x)\bar{i}$, where x is in meters along vertical direction and velocity is in m/s. Coordinate x is zero at the start of the pipe section and positive downwards. Viscous effects are neglected. Determine the pressure at the end of the pipe as a function of x . If the length of the pipe is 10 m, what is the pressure at the end? If the pressure at the starting point is 70 KPa, what is the pressure at the end?

[6]